



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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Frequentist View

- *With Prior Knowledge*: out of all n different but equally likely possible outcomes of an event, there are k that have a property A which is therefore measured with probability

$$P(A) = \frac{k}{n}$$

- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

$$P(A) = \lim_{n \rightarrow \infty} \frac{k}{n}$$

Bayesian View

- $P(A|B)$ quantifies the plausibility of an assumption A under known information given by B
- *Kolmogorov Axioms* underpin the rules for calculating with probabilities
- $P(A, B) = P(A) P(B|A)$ quantifies the probability of measuring A and B simultaneously, and from assuming $P(A, B) = P(B, A)$ we find

Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Continuous Distributions

■ *Cumulative Distribution Function:* $F(x) = P(X \leq x)$

■ *Probability Density Function:* $f(x) = \frac{d}{dx}F(x)$

$$P(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$$

■ *Normalization:* $1 = \int_{-\infty}^{+\infty} f(x) dx \quad \lim_{x \rightarrow -\infty} F(x) = 0 \quad \lim_{x \rightarrow +\infty} F(x) = 1$

Expectation Values

$$E[h(x)] = \int_{-\infty}^{+\infty} h(x) f(x) dx$$

■ *Mean* or first raw moment: $\mu = E[X] = \int_{-\infty}^{+\infty} x f(x) dx$

■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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Drawing Numbers

$$x \sim U(a, b)$$

$$y \sim N(\mu, \sigma)$$

■ *Uniform Distribution:* $U(a, b) = f(x|a, b) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$

■ *Gaussian Distribution:* $N(\mu, \sigma) = f(y|\mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \left(\frac{y-\mu}{\sigma}\right)^2\right)$

Number Generation

- *Pseudo Random Numbers: $x_{j+1} = g(x_1, \dots, x_j)$*
 - deterministic & reproducible
 - efficient & transparent
 - almost indistinguishable with long periods of recurrence
- *Equal Distribution Generators:*
 - *Linear Congruential*
 - *Linear Feedback Shift*
 - *Permuted Congruential*
- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

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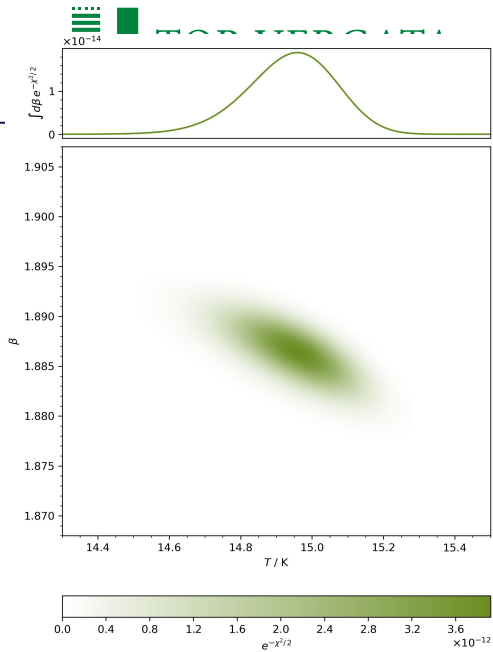
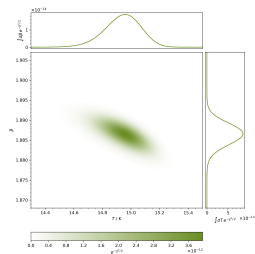
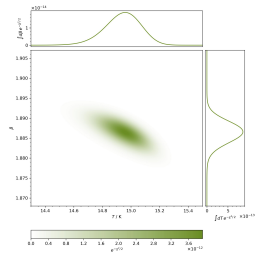
Markov Chain Monte Carlo

Popular Algorithms

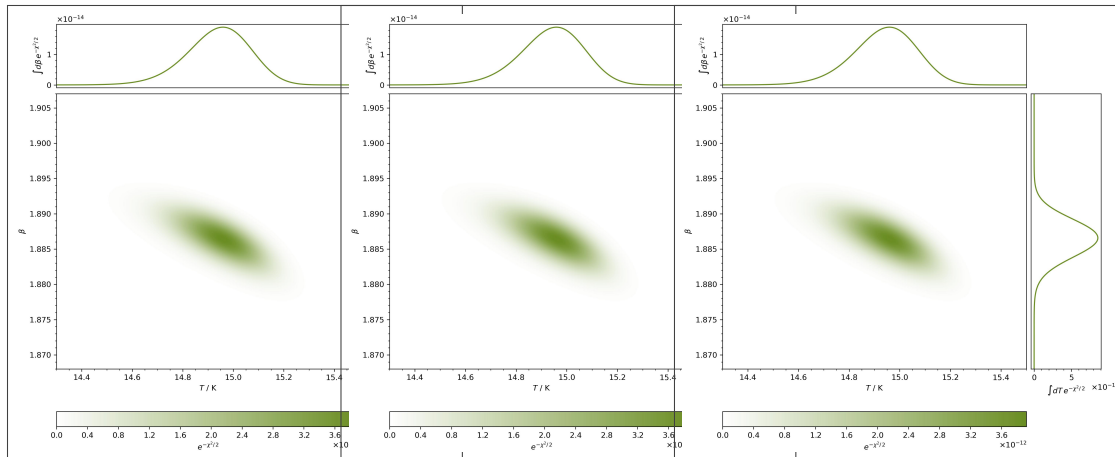
Tuning & Diagnosing

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Where I Studied

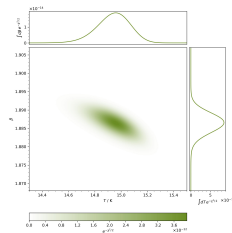
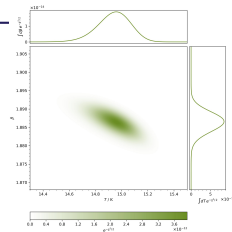
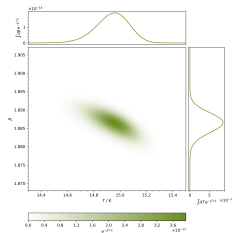
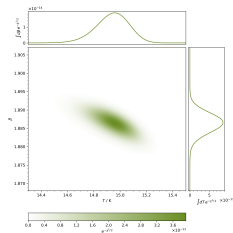


What I Studied

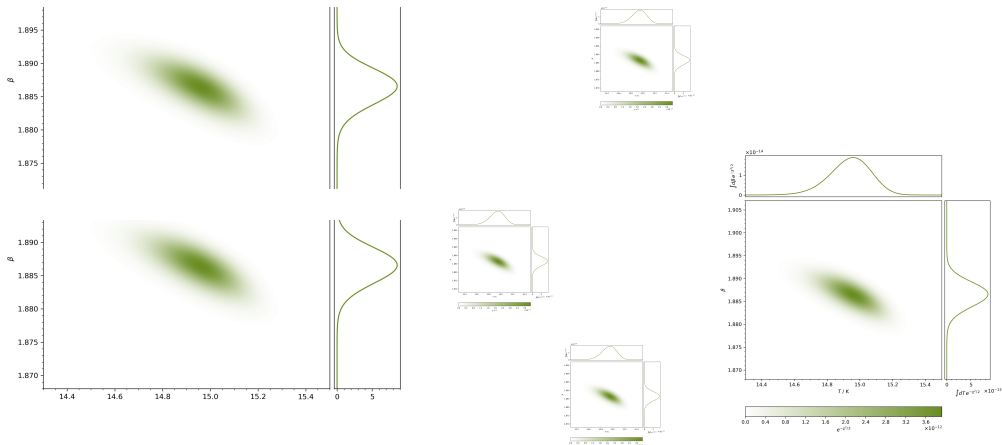


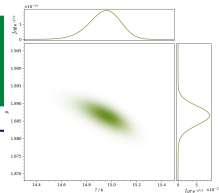
Why Astrophysics

- Interesting
 - Awesome Scales
 - Readily Observable
- Interdisciplinary
- International



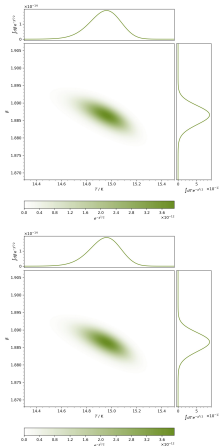
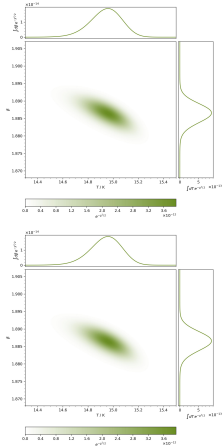
Where I Have Been





Other Interests

- Nature Photography
- Hiking & Cycling
- Programming
- Politics
- Meeting New People



Optimisation

Thank You!